Final Review

ERIN KEITH

Goals

1. Where we've been
2. Where you're going

Additional Announcement

Evaluations are available through Canvas

- For every 25%, **I will add 1 point to all final exam grades**.
- Currently at 20% responses

Constructive criticism can follow a "plus, minus, delta" format, where "plus" is something good, "minus" is something that needs work, and "delta" is something to change (even if it's adding more of a good thing!).

Please convey your feedback as respectfully as possible.

Final Exam

- Tuesday, December 16^{th} from 3pm – 5pm
- Online test in classroom
  - Log into the lab computers
- No hats or sunglasses

Where we've been

Introduction to design patterns: Strategy, Observer, Factory, Singleton, Command, Adapter, Facade, Template Method, Iterator, Composite, State, Proxy. Object Oriented design principles, Java/UML.

Where we've been

IMPROVED SKILLS

- Object Oriented Design and UMLs
- Java Programming

NEW INFORMATION

- Strategy
- Observer
- Decorator
- Factory Method
- Abstract Factory
- **Singleton**
- **Command**
- **Adapter**
- **Composite**
- **State**
- **MVC**

Final Structure

Topic	Points Possible	Points Received
Identification	10	
Analysis	30	
Design	60	
Total	**100**	

Types of Patterns

- Behavioral Patterns – interaction or communication between objects
- Creational patterns - object creation
- Structural patterns - object and class composition

Behavioral Patterns

- Behavioral design patterns are concerned with **the interaction and responsibility of objects.**
- In these design patterns, **the interaction between the objects should be in such a way that they can easily talk to each other and still should be loosely coupled.**
- These patterns increase flexibility

Identification

Know the difference between the different categories of patterns (structural, creational, behavioral) and which pattern belongs to which category.

Analysis

Be able to compare and contrast (discuss similarities and differences) between any two patterns.

Be able to draw generic UML diagrams for any design pattern.

Design

You will be given 3 - 4 design problems.

- Be able to select the most appropriate design pattern to use and provide the pattern's *general* intent.
- Clearly describe how this pattern *specifically* addresses the problem.
- *Explain* how this pattern should be implemented.
  - no code!
- Draw a UML class diagram to illustrate the implementation of your pattern *for this problem*.
  - Provide a UML specifically for the design problem not a generic class diagram for the design pattern.

Where we've been

- You have come so far over the years!
- It has been my privilege to work with you and watch you grow.
- I am proud of you.

Thank you so much!

If you ever need anything, please ask!

- Career or school advice
- Recommendations
- Help with other classes
- Swing by my office! WPEB 433

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Teaching you the CS ***why's*** so you can be CS ***wise***.

Next Time

Final

- Tuesday, December 16th from 3pm – 5pm
- Online test in classroom
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